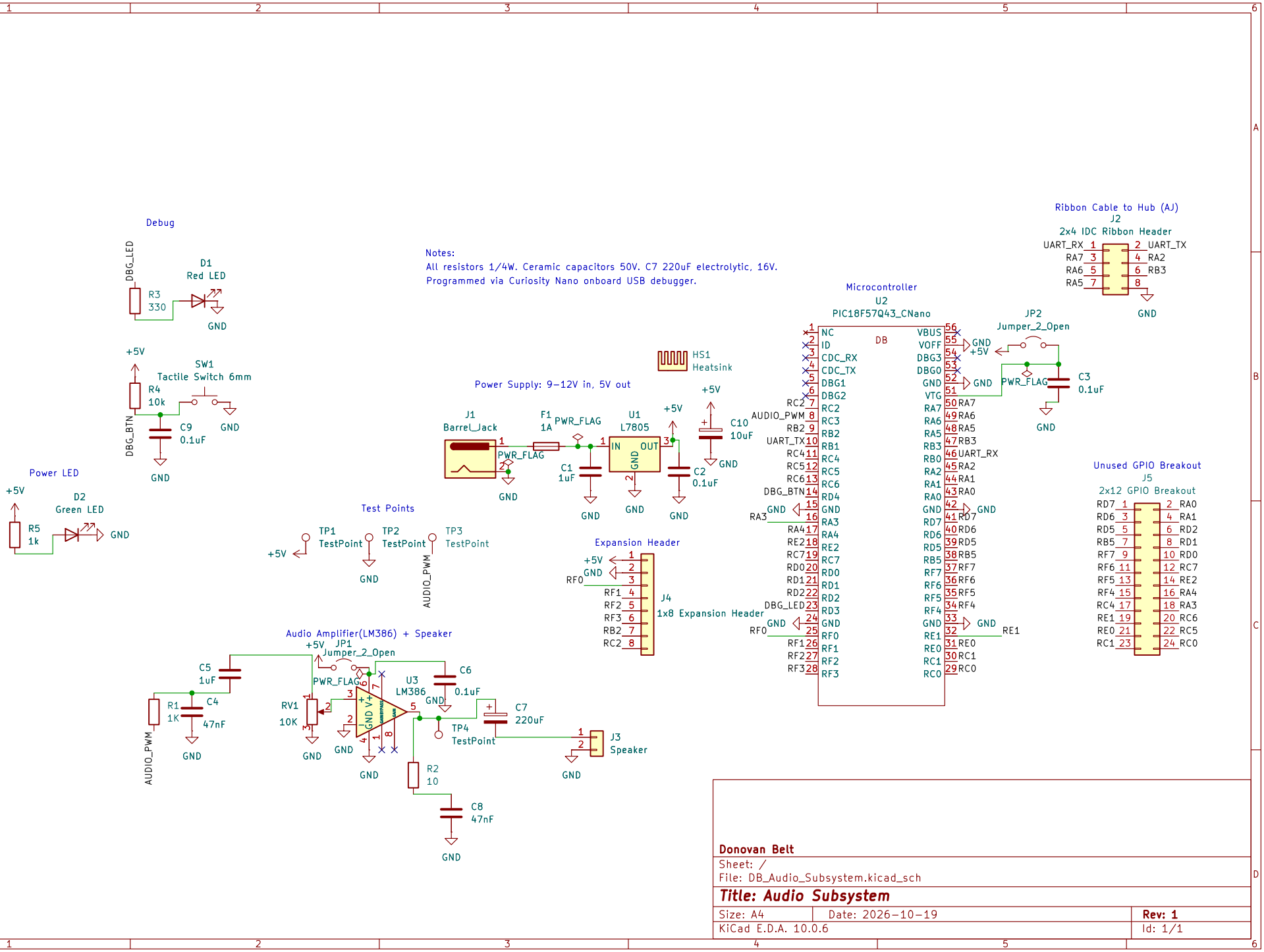

Debug
D1
Red LED
DBG_LED
R3
330
GND
+5V
SW1
Tactile Switch 6mm
R4
10k
DBG_BTN
C9
0.1uF
GND
Power LED
+5V
D2
Green LED
R5
1k
GND
Notes:
All resistors 1/4W. Ceramic capacitors 50V. C7 220uF electrolytic, 16V.
Programmed via Curiosity Nano onboard USB debugger.
Power Supply: 9–12V in, 5V out
J1
Barrel_Jack
F1
1A
PWR_FLAG
U1
L7805
IN
OUT
GND
C1
1uF
C2
0.1uF
C10
10uF
+5V
HS1
Heatsink
Test Points
TP1
TP2
TP3
TestPoint
AUDIO_PWM
Expansion Header
J4
1x8 Expansion Header
Audio Amplifier(LM386) + Speaker
JP1
Jumper_2_Open
U3
LM386
C5
1uF
R1
1K
C4
47nF
RV1
10K
C6
0.1uF
C7
220uF
TP4
J3
Speaker
R2
10
C8
47nF
Microcontroller
U2
PIC18F57Q43_CNano
DB
JP2
Jumper_2_Open
C3
0.1uF
Ribbon Cable to Hub (AJ)
J2
2x4 IDC Ribbon Header
UART_RX
UART_TX
RA7
RA6
RA5
RA2
RB3
Unused GPIO Breakout
J5
2x12 GPIO Breakout
Donovan Belt
Sheet: /
File: DB_Audio_Subsystem.kicad_sch
Title: Audio Subsystem
Size: A4
Date: 2026-10-19
Rev: 1
KiCad E.D.A. 10.0.6
Id: 1/1